Which Virtualization Should YOU Choose?

Quick Decision Tree

```
START HERE
│
├─ Do you have dedicated hardware for hypervisor?
│  ├─ YES → Proxmox (best management, testing)
│  └─ NO → Continue...
│
├─ What matters MOST to you?
│  ├─ Fastest boot time → Firecracker
│  ├─ Best GUI quality → QEMU MicroVM
│  ├─ Easiest setup → QEMU MicroVM
│  └─ Modern + balanced → Cloud Hypervisor
│
└─ For your FINAL PROJECT → Cloud Hypervisor
   (Best alignment with PID goals)
```

Detailed Comparison Matrix

Criterion	Firecracker	Cloud Hypervisor	QEMU MicroVM	Proxmox
Boot Time	🥇 125ms	🥈 100-150ms	🥉 400-600ms	30-60s
GUI Setup	❌ Complex (VNC)	⚠️ Medium (VNC/ virtio-gpu)	✅ Easy (GTK/ SDL)	✅ Easy (SPICE)
Display Quality	⭐⭐ (VNC lag)	⭐⭐⭐ (Good)	⭐⭐⭐⭐⭐ (Native)	⭐⭐⭐⭐⭐ (SPICE)
Setup Difficulty	😰 Hard	😰 Hard	😊 Medium	😊 Medium
Hardware Isolation	✅ Yes (own kernel)	✅ Yes (own kernel)	✅ Yes (own kernel)	✅ Yes
GPU Acceleration	❌ No	✅ Yes (virtio-gpu)	✅ Yes	✅ Yes
Memory Overhead	🥇 5MB	🥈 10MB	🥉 50MB	500MB+
Documentation	⭐⭐⭐	⭐⭐	⭐⭐⭐⭐⭐	⭐⭐⭐⭐⭐
Community Size	Large (AWS)	Small	Huge	Large
PID Alignment	✅ Perfect	✅ Perfect	✅ Perfect	✅ Perfect
Novelty Factor	⭐⭐⭐⭐	⭐⭐⭐⭐⭐	⭐⭐⭐	⭐⭐

For YOUR Project - Recommendations

🏆 BEST CHOICE: Cloud Hypervisor

Why:

1. ✅ Own kernel (hardware isolation) - aligns with PID
2. ✅ GPU/GUI support - solves display problem
3. ✅ Fast boot (100-150ms) - competitive performance
4. ✅ Novel - no one else using it for browser isolation
5. ✅ Modern Rust-based - good for academic credibility
6. ✅ Active development - shows cutting-edge knowledge

Use this for: Final April submission

🥈 RUNNER-UP: QEMU MicroVM

Why:

1. ✅ Easiest GUI setup (GTK/SDL native)
2. ✅ Best documentation and support
3. ✅ Mature and stable
4. ✅ Own kernel (hardware isolation)
5. ✅ GPU acceleration works perfectly

Use this if: Cloud Hypervisor setup fails, or you need something working FAST

🥉 FOR TESTING: Proxmox

Why:

1. ✅ Perfect for comparing multiple configurations
2. ✅ Easy to create snapshots and test different setups
3. ✅ Can clone VMs quickly to test Firecracker vs QEMU vs Cloud Hypervisor
4. ✅ Web UI makes testing easier

Use this for: Initial experimentation and comparison

⚡ FOR SPEED DEMOS: Firecracker

Why:

1. ✅ Fastest boot (125ms) - impressive for demos
2. ✅ AWS-proven technology - strong credibility
3. ✅ Can cite extensive research papers
4. ❌ But VNC setup is painful

Use this for: Benchmarking boot time only, or if you want to show "AWS technology"

Testing Strategy (RECOMMENDED)

Week 1: Setup All Four

Day 1-2: Proxmox

- Install on spare machine or nested VM
- Create browser isolation VM template
- Test SPICE display
- Take screenshots

Day 3: QEMU MicroVM

- Easiest to set up
- Test GTK display mode
- Measure boot time
- Take screenshots

Day 4: Cloud Hypervisor

- Build from source
- Test with VNC
- Measure boot time
- Take screenshots

Day 5: Firecracker

- Most complex setup
- Test VNC streaming
- Measure boot time
- Document challenges

Weekend: Compare and Decide

What to Include in Your Report

For February Prototype Demo:

Show TWO options:

1. **QEMU MicroVM (working demo)**
    - "This demonstrates the concept with mature technology"
    - Show actual browser running
    - Easy to get working
2. **Cloud Hypervisor (in progress)**
    - "This is the final target for superior performance"
    - Show boot time comparisons
    - Explain why it's better

For April Final Submission:

Primary: Cloud Hypervisor

- "Selected for optimal balance of security, performance, and modern architecture"
- Boot time: 100-150ms
- GPU support via virtio-gpu
- Rust-based implementation for memory safety

Backup: QEMU MicroVM

- If Cloud Hypervisor gives problems
- Still meets all requirements
- More documentation available

Effort vs Reward Analysis

```
High Reward
│
│   Cloud Hypervisor ★
│       │
│       │  Firecracker
│       │     │
│   QEMU MicroVM
│       │
│   Proxmox
│
└─────────────────── High Effort
    Low


★ = Sweet spot for your project
```

Cloud Hypervisor = Best effort/reward ratio

- Hard enough to be impressive
- Not so hard you'll fail
- Novel enough for academic contribution
- Practical enough to actually work

Timeline Allocation

Total Time: 40 hours over 2 weeks

Option	Setup	Config	Testing	Total
Proxmox	4h	2h	2h	8h
QEMU	3h	2h	2h	7h
Cloud Hypervisor	6h	4h	3h	13h
Firecracker	8h	5h	4h	17h
Total				**45h**

Allows buffer time for issues

My Final Recommendation

Do This (In Order):

1. **Today: Start with QEMU MicroVM**
   - Follow guide above
   - Get browser running in 3-4 hours
   - Take screenshots for interim report
   - ✅ You now have a working prototype!
2. **Tomorrow: Set up Proxmox (if you have spare hardware)**
   - Use it to test other options
   - Create VM templates
   - Easy to snapshot/rollback
3. **This Week: Build Cloud Hypervisor**
   - This is your MAIN platform
   - Spend time getting it right
   - Document everything
4. **Next Week: Try Firecracker (optional)**
   - Only if you have time
   - Good for boot time comparisons
   - Can mention "evaluated but chose Cloud Hypervisor for GUI support"

For Your Report:

Introduction:

> "Multiple virtualization platforms were evaluated including Firecracker microVMs (AWS), Cloud Hypervisor, QEMU microVM mode, and Proxmox VE. Initial prototyping utilized QEMU microVM for rapid development, with production implementation targeting Cloud Hypervisor for optimal balance of security (hardware-level isolation), performance (100-150ms boot time), and functionality (GPU support via virtio-gpu)."

This shows:

- You did thorough research ✅
- You tested multiple options ✅
- You made informed decision ✅
- You're technically capable ✅

Summary: Just Tell Me What to Do!

For February Demo (3 weeks from now):

1. Use **QEMU MicroVM**
2. Get it working this week
3. Take screenshots
4. Add to interim report

For April Final (2 months from now):

1. Use **Cloud Hypervisor**
2. Build it properly over March
3. Compare performance with QEMU
4. Write about why you chose it

If Everything Fails:

1. Fall back to **QEMU MicroVM**
2. It's still hardware isolation (own kernel)
3. Still meets all PID requirements
4. Professor will be happy

Don't overthink it! Start with QEMU today, get something working, then gradually move to Cloud Hypervisor.

You'll have 4 guides to follow - you can't fail! 🚀